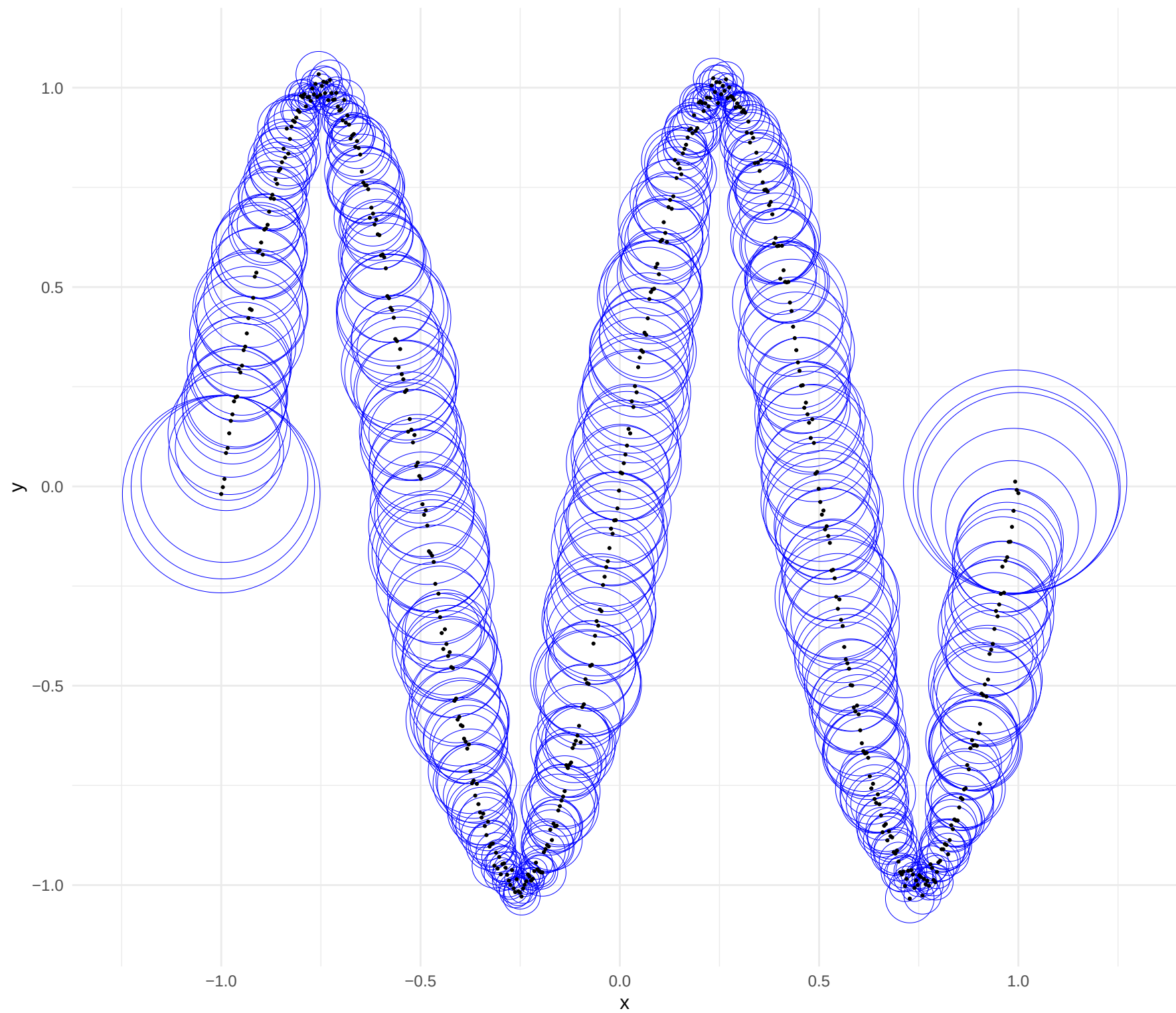
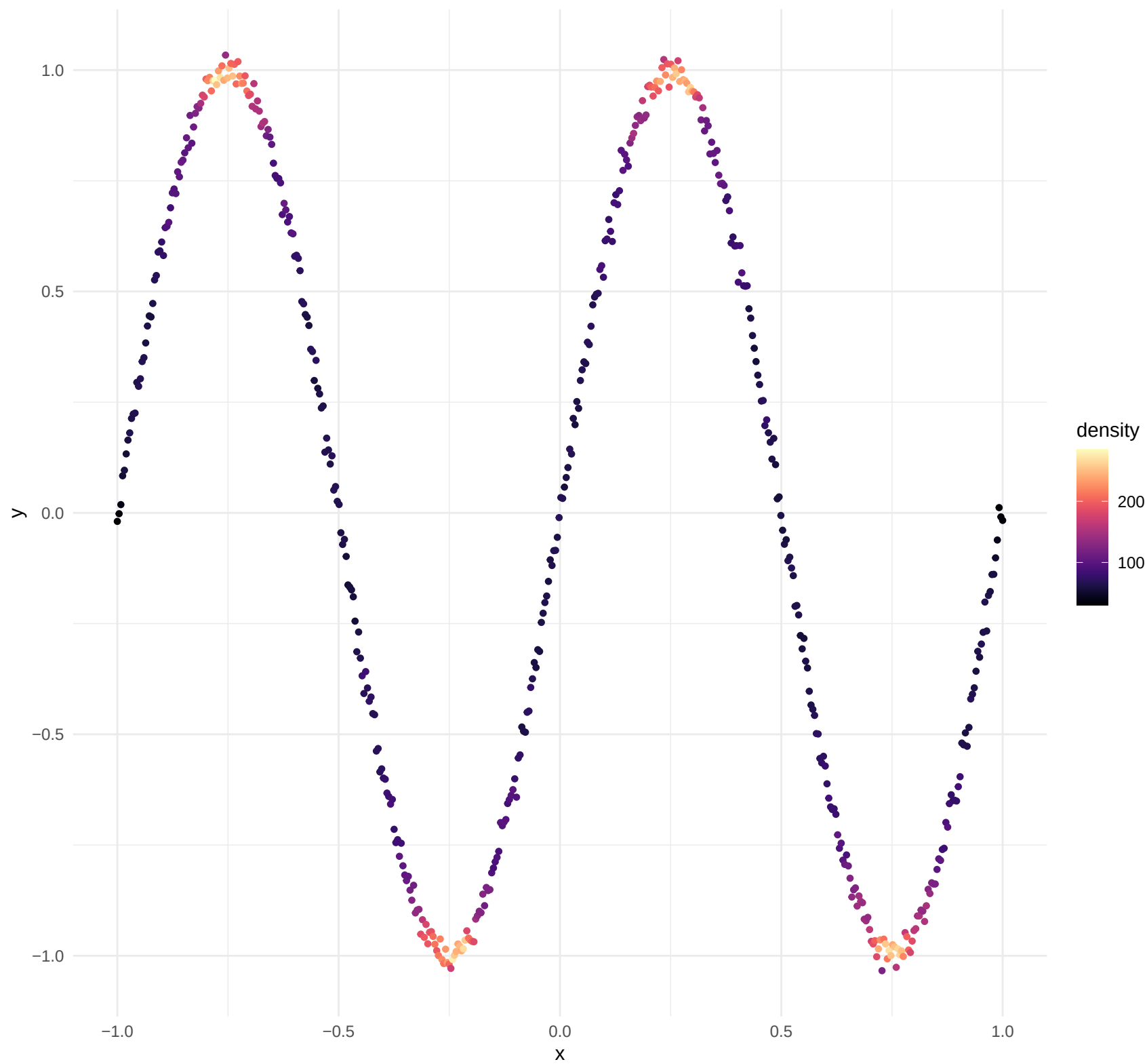


All Adaptive Spheres

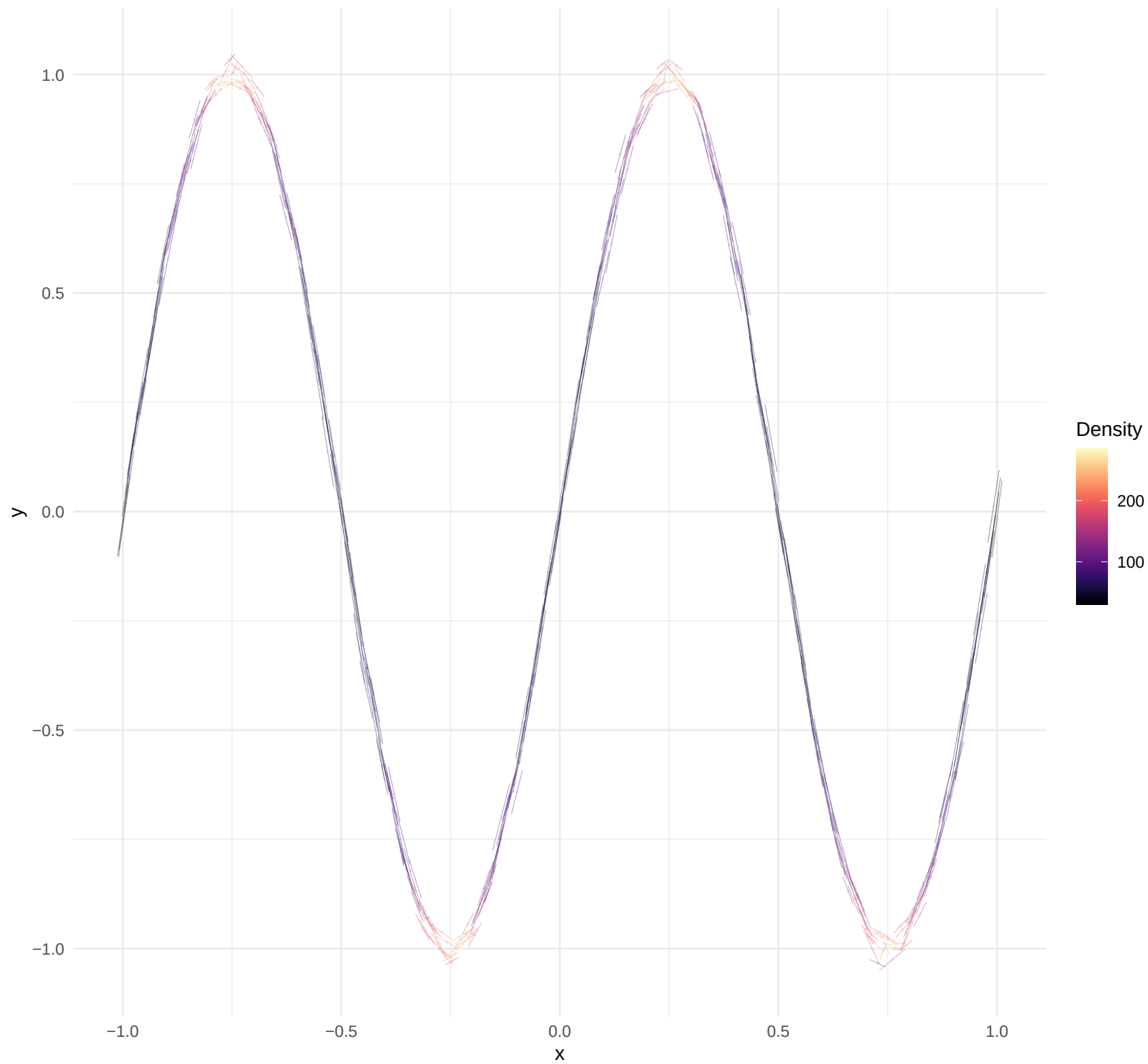
Visualizing local neighborhood size (radius) for every point.



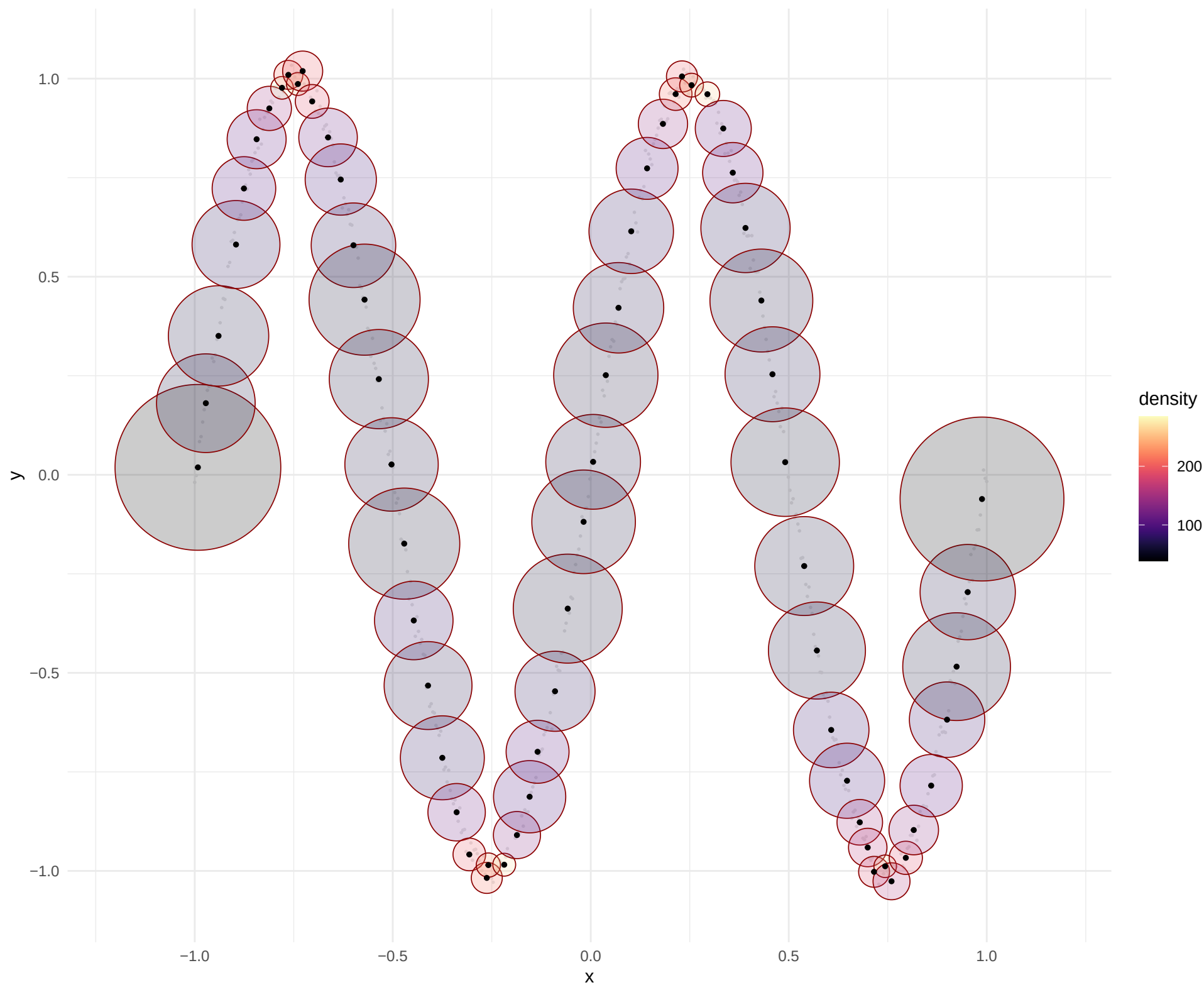
Point Density Distribution



Global Tangent Field (PC1)

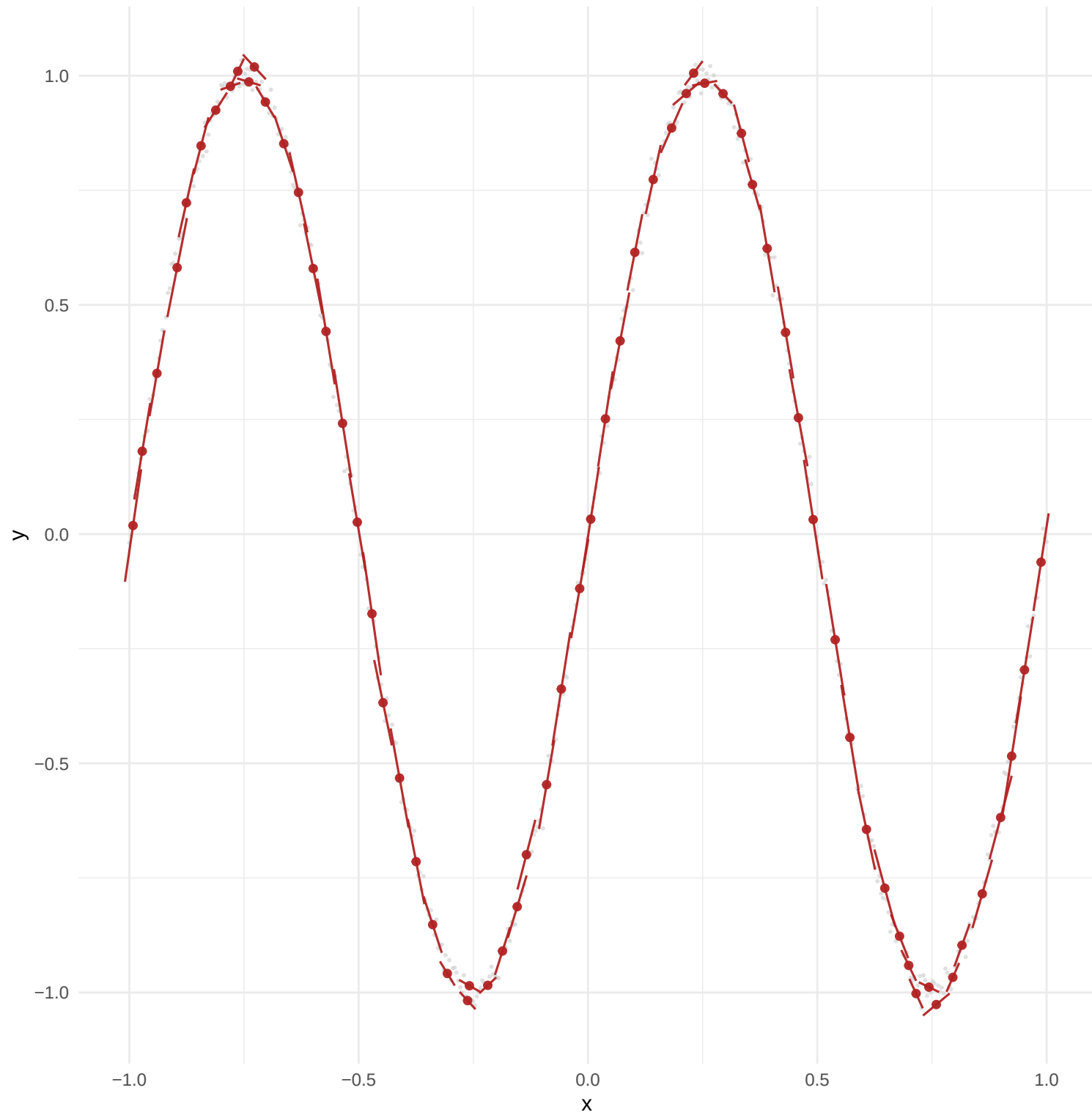


Atlas Selection (Anchors)



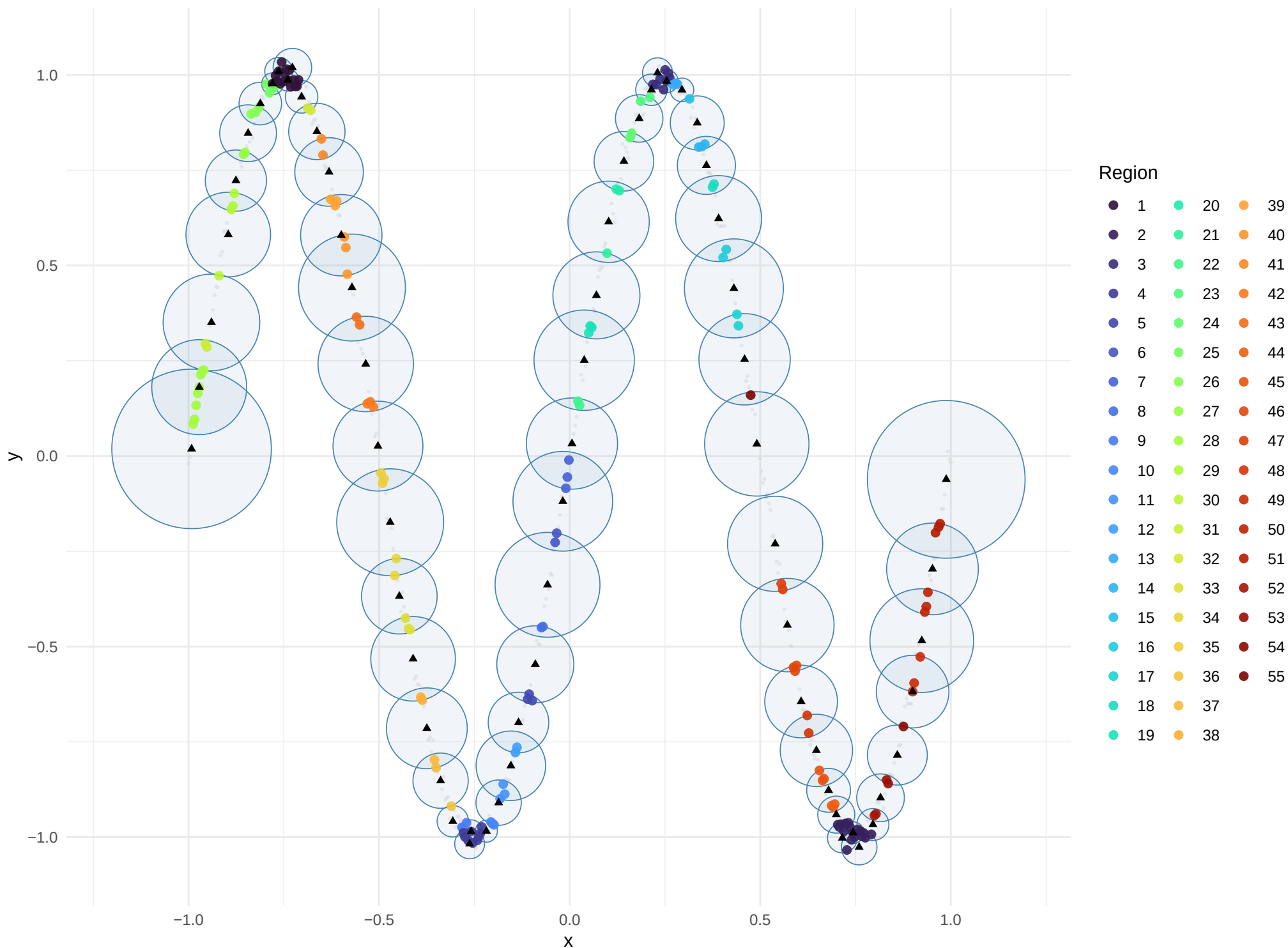
Anchor SVD Segments

Same anchors as plot 4: spheres first, then their SVD (PC1) in red



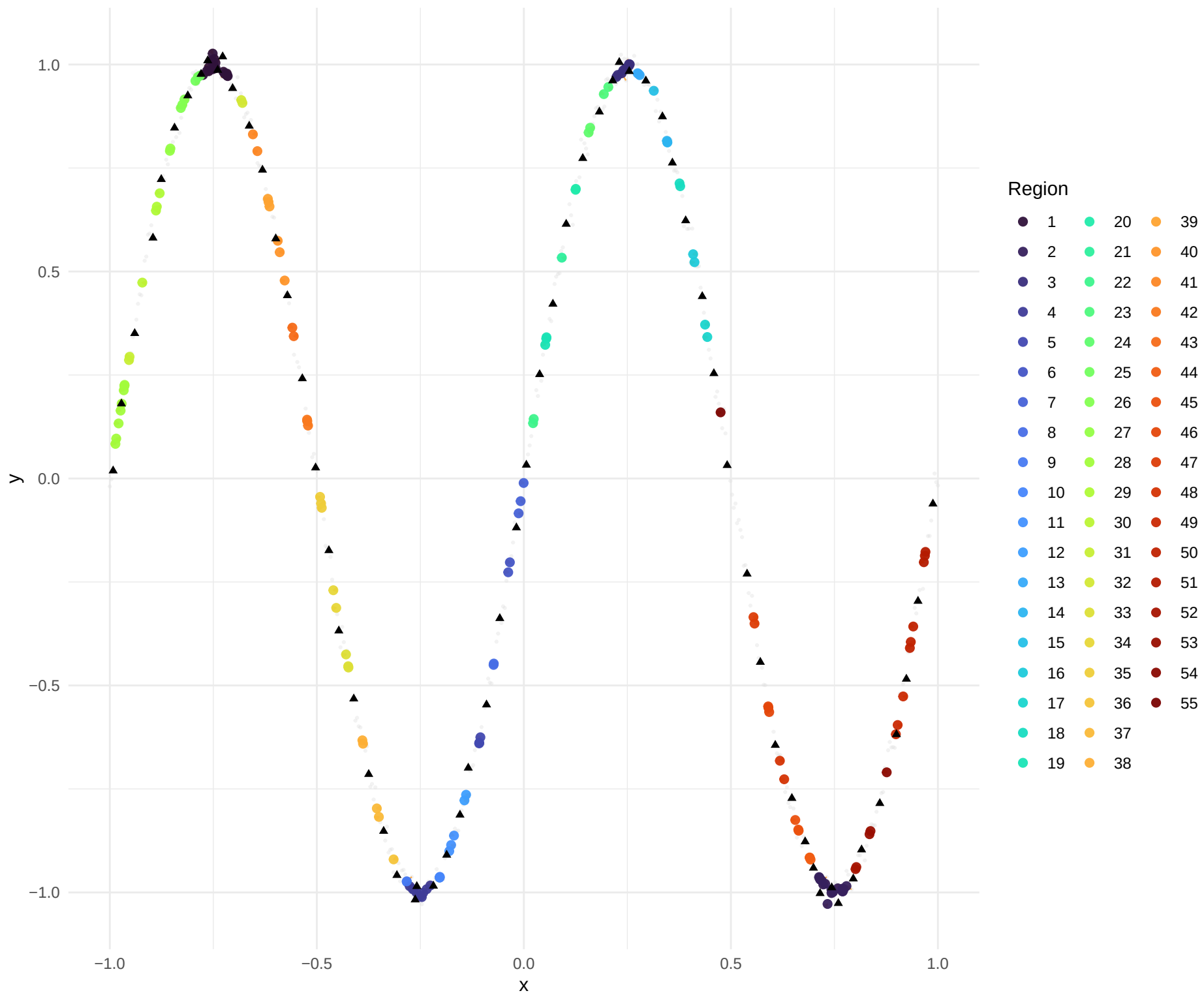
Intersection Regions (2+ anchors)

Anchor spheres in blue. Colored points = overlap zones. Triangles = anchor centers.



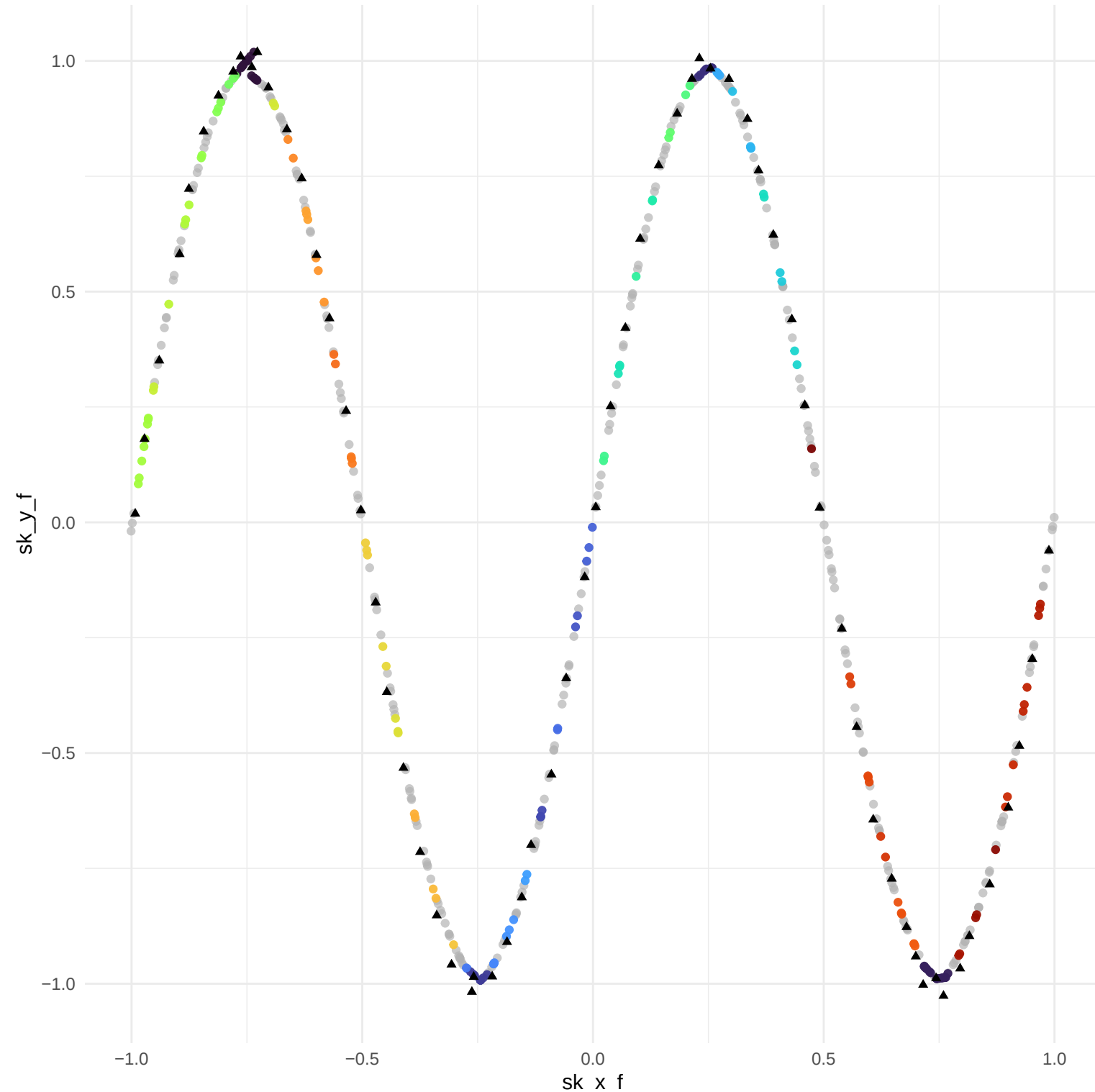
Stitched Backbone (Iteration 1)

Orange path: stitching; Triangles: original anchors



Final Stitched Backbone (Converged)

Final stitched points



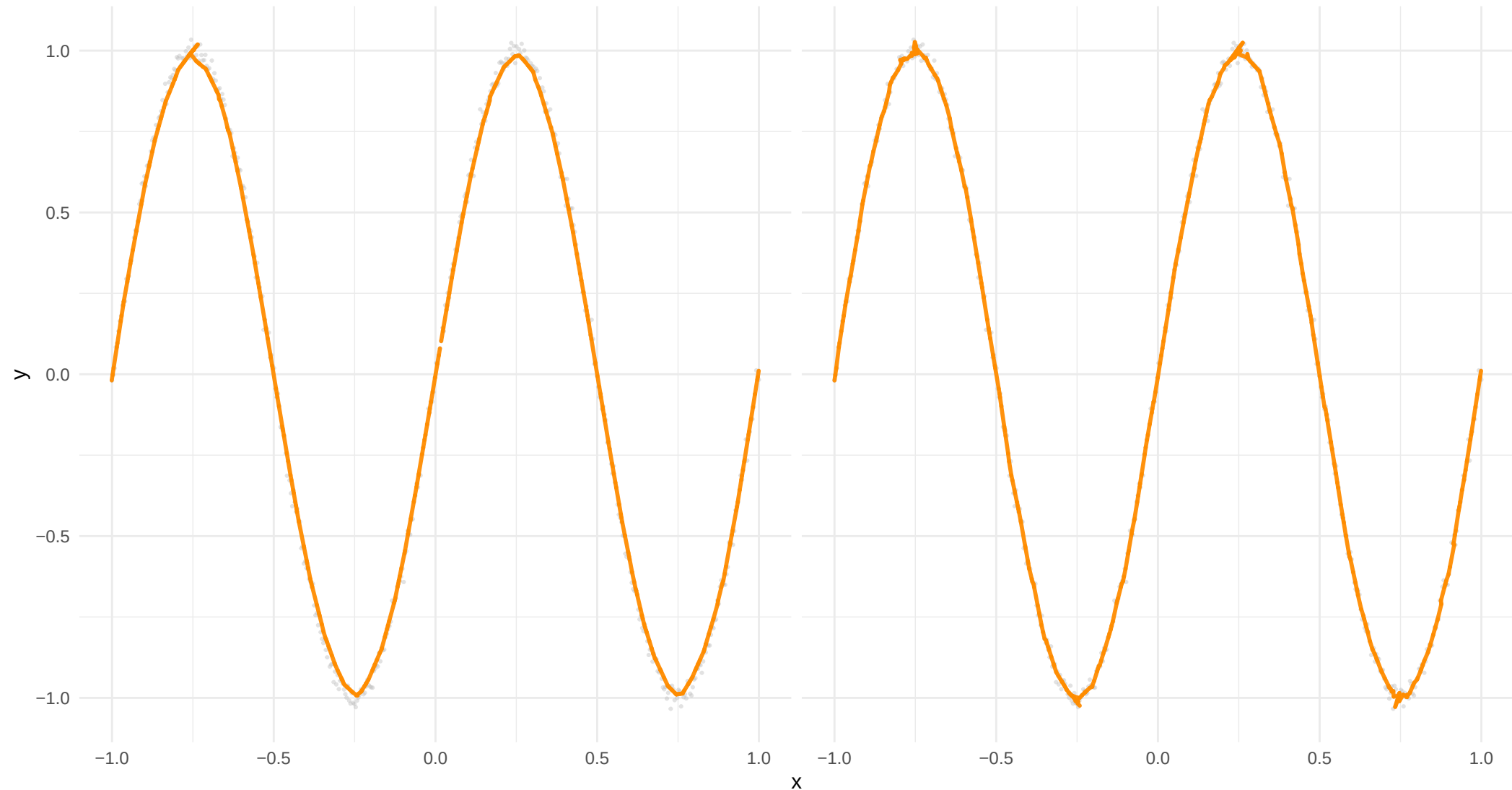
Region

- | | | |
|----|----|----|
| 1 | 20 | 39 |
| 2 | 21 | 40 |
| 3 | 22 | 41 |
| 4 | 23 | 42 |
| 5 | 24 | 43 |
| 6 | 25 | 44 |
| 7 | 26 | 45 |
| 8 | 27 | 46 |
| 9 | 28 | 47 |
| 10 | 29 | 48 |
| 11 | 30 | 49 |
| 12 | 31 | 50 |
| 13 | 32 | 51 |
| 14 | 33 | 52 |
| 15 | 34 | 53 |
| 16 | 35 | 54 |
| 17 | 36 | 55 |
| 18 | 37 | |
| 19 | 38 | |

Backbone Edges

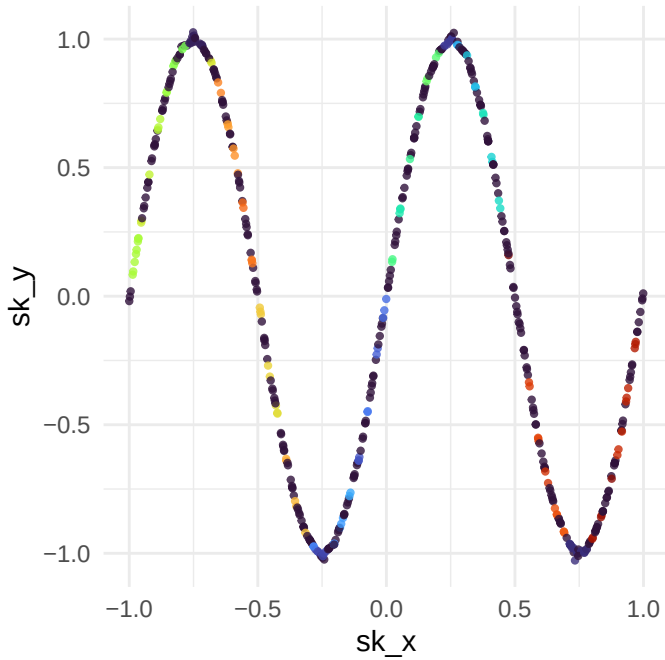
Final

Iteration 1

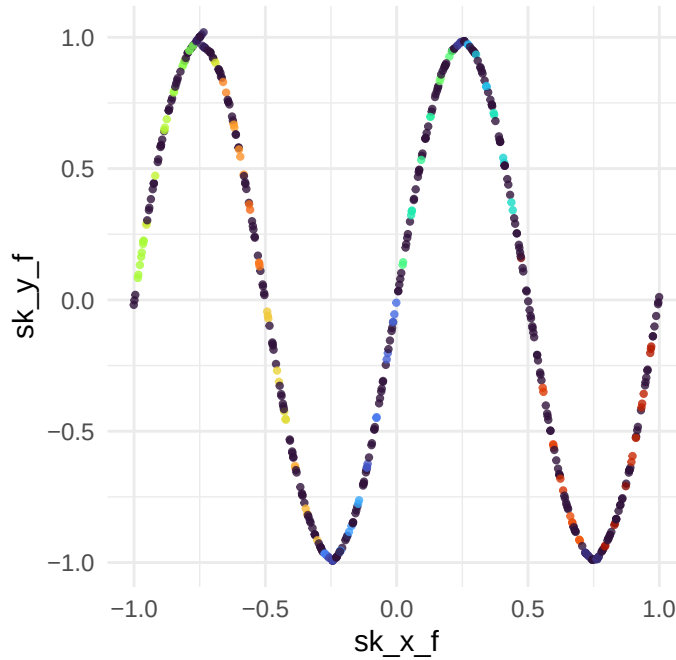


Convergence: iteration 1 vs final

Iteration 1

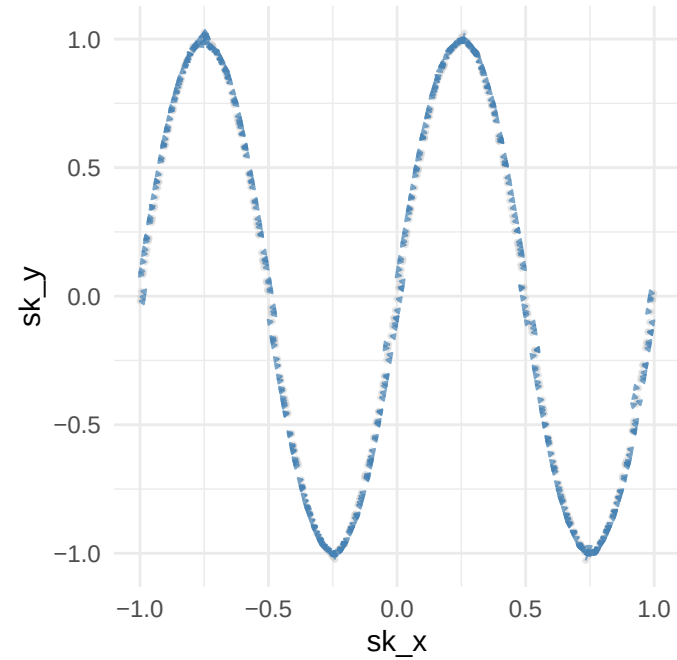


Converged



Displacement iter1 -> final

Points moved > 1e-6: 500 / 500



Parameters: $k_{min} = 10$, $g_{threshold} = 3.0$, $o_{max} = 0.3$, $o_{min} = 0.10$, $stability = 0.90$, $scale_factor = 2.5$, $max_iter = 50$, $conv_tol = 0.01$